

LAS VEGAS MCCARRAN, NV, USA

WMO: 723860

Lat: 36.072N

Lon: 115.163W

Elev: 2180

StdP: 13.57

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23169

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 32.8	(c) 35.4	(d) -1.4	(e) 5.5	(f) 60.0	(g) 2.0	(h) 6.6	(i) 56.4	(j) 27.4	(k) 50.9	(l) 24.7	(m) 50.8	(n) 4.9	(o) 200	(p) 0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.5	(c) 109.0	(d) 67.0	(e) 106.7	(f) 66.4	(g) 104.6	(h) 65.7	(i) 72.6	(j) 96.2	(k) 71.2	(l) 94.6	(m) 69.9	(n) 94.2	(o) 9.9	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	103.6	82.0	63.4	94.7	84.2	60.7	86.0	87.3	37.6	96.2	36.3	94.3	35.2	93.8	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.1	(b) 21.9	(c) 19.2	(e) 28.3	(f) 112.8	(g) 3.1	(h) 2.1	(i) 26.0	(j) 114.3	(k) 24.2	(l) 115.5	(m) 22.4	(n) 116.7	(o) 20.2	(p) 118.2		
(a) 25.1	(b) 21.9	(c) 19.2	(e) 22.0	(f) 74.8	(g) 2.7	(h) 2.2	(i) 20.0	(j) 76.4	(k) 18.5	(l) 77.6	(m) 17.0	(n) 78.9	(o) 15.0	(p) 80.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.0	49.6	53.1	61.0	67.4	77.0	87.8	93.2	91.4	83.3	69.9	57.3	48.3
	DBStd	16.98	5.45	6.38	6.97	7.65	7.91	6.82	4.69	4.30	6.02	7.05	6.88	5.47
	HDD50	232	74	37	5	1	0	0	0	0	0	0	17	98
	HDD65	1841	479	333	162	63	8	1	0	0	0	34	242	519
	CDD50	7546	60	125	346	523	837	1134	1338	1285	1000	617	237	44
	CDD65	3681	0	1	39	136	380	684	873	820	550	186	12	0
	CDH74	54707	0	15	379	1415	4872	10837	14670	13264	7408	1752	95	0
	CDH80	34379	0	0	74	505	2552	7109	10313	8968	4235	618	5	0
Wind	WSAvg	8.0	6.3	7.2	8.7	10.1	9.8	9.9	8.6	8.3	7.5	7.1	6.3	6.2
Precipitation	PrecAvg	4.2	0.5	0.6	0.4	0.2	0.2	0.0	0.4	0.4	0.3	0.3	0.4	0.4
	PrecMax	9.8	3.0	2.9	4.8	2.4	0.9	0.5	2.5	2.1	2.0	1.5	2.2	2.1
	PrecMin	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.0	0.7	0.8	0.7	0.4	0.2	0.1	0.5	0.5	0.5	0.4	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.9	76.5	86.7	94.0	103.1	111.2	112.8	109.4	104.2	93.8	80.5	68.1
		MCWB	49.5	51.9	55.7	59.2	65.1	64.2	68.4	68.4	65.9	61.8	54.2	50.3
	2%	DB	65.5	72.4	81.6	89.4	98.9	107.3	110.0	107.3	101.0	89.9	76.8	64.3
		MCWB	47.9	50.3	54.1	57.0	61.3	63.8	67.5	67.3	64.4	59.9	53.0	48.1
	5%	DB	62.9	68.8	78.2	85.9	95.8	104.8	107.7	105.2	98.7	86.5	73.5	61.5
		MCWB	46.4	48.7	52.4	55.6	59.9	63.6	67.4	66.5	63.9	58.1	51.6	46.4
	10%	DB	60.2	65.3	74.9	82.2	92.3	102.1	105.3	103.0	96.3	83.2	70.0	58.5
		MCWB	45.3	47.4	51.0	54.0	58.7	62.9	66.5	65.8	63.3	56.6	50.5	44.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	55.4	57.7	61.2	67.5	72.2	75.0	74.3	72.4	65.5	58.1	53.4
		MCDB	61.9	68.7	78.7	90.4	99.8	102.6	101.0	99.3	89.3	83.8	72.0	63.2
	2%	WB	51.1	52.7	55.2	58.5	63.8	68.4	72.9	72.3	69.6	62.6	55.0	50.9
		MCDB	60.3	65.7	77.1	85.2	94.0	101.0	96.5	96.2	89.5	84.1	72.8	60.3
	5%	WB	49.2	50.9	53.6	56.7	61.5	66.3	71.5	71.0	68.2	60.3	52.9	48.7
		MCDB	59.0	64.4	74.9	82.6	91.1	99.5	94.5	94.6	89.1	81.1	70.9	58.2
	10%	WB	47.2	49.0	52.0	54.9	59.6	64.3	70.3	69.7	66.5	58.6	51.2	46.4
		MCDB	57.0	62.4	71.9	79.5	89.3	97.9	94.9	93.4	88.9	78.9	68.0	56.5
Mean Daily Temperature Range	5% DB	MDBR	17.8	18.5	20.5	20.9	21.7	22.9	21.5	21.3	21.5	20.7	19.2	16.7
		MCDBR	21.2	23.3	24.6	25.3	24.8	25.0	23.7	23.0	23.5	24.0	22.8	20.3
		MCWBR	11.0	11.0	10.6	10.8	9.9	9.6	9.3	8.9	9.5	10.5	10.9	10.8
	5% WB	MCDBR	17.0	19.6	23.2	23.8	24.2	24.1	20.3	20.3	20.8	21.6	20.9	16.4
		MCWBR	9.6	9.9	10.4	10.4	11.2	11.9	8.3	8.7	8.9	10.1	10.6	9.4
Clear-Sky Solar Irradiance	taub	0.264	0.270	0.283	0.305	0.323	0.319	0.359	0.349	0.314	0.291	0.268	0.264	
	taud	2.618	2.620	2.557	2.474	2.447	2.469	2.443	2.461	2.538	2.591	2.625	2.627	
	Ebn at Noon	303	313	315	310	304	304	291	292	299	300	299	296	
	Edh at Noon	23	26	30	34	36	35	36	34	30	26	23	21	
All-Sky Solar Radiation	RadAvg	976	1281	1758	2204	2497	2707	2385	2218	1959	1475	1100	865	
	RadStd	77	112	81	92	122	110	125	112	81	86	45	63	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+1.10	N/A	-2.68	N/A	N/A	N/A	N/A	-196	+396	+216
Regional (15 neighbors)	+0.79	N/A	-2.88	N/A	N/A	N/A	N/A	N/A	+286	N/A

Nomenclature: See separate page